

Set theory short proofs

2.1 Countable sets

TH 2.1 : (AC) Finite products and countable unions of countable sets are countable (Cantor). [AC : an infinite product of arbitrary countable sets is not countable ; for $\mathbb{N}^{\mathbb{N}}$ use diagonal argument.]

(1) Let $A_{1 \leq i \leq n}$ be countable sets (each) with a countable maps c_i .

(2) Consider the map $C : \prod_{i=1}^n A_i \mapsto \mathbb{N}$ given by $C((a_1, \dots, a_n)) = \prod_{i=1}^n p_i^{c_i(a_i)}$ where p_i are different primes.

(3) Since prime factorization is unique, this map is injective.

(4) But each infinite subset of the natural numbers is countable, hence the product is countable as well.

(5) Now consider an infinite family of countable sets A_i for i natural with counting maps c_i .

(6) We have $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \{c_i^{-1}(n) \mid n \in \mathbb{N}\} = \{c_i^{-1}(n) \mid (n, i) \in \mathbb{N}^2\}$.

(7) This set is countable since $\mathbb{N}^2$ is countable. □

TH 2.2 : The integers $\mathbb{Z}$, the rationales $\mathbb{Q}$ and the algebraic numbers $\mathbb{A}$ are countable.

(1) The integers are the union of positive and negative numbers, two countable sets, hence they are countable.

(2) The map $f : \mathbb{Q} \mapsto \mathbb{Z}^2$; $f(p/q) = (p, q)$ is injective and $\mathbb{Z}^2$ is countable, hence $\mathbb{Q}$ is countable as well.

(3) There are countable many polynomials of degree n over $\mathbb{Q}$ since $\mathbb{Q}^n$ is countable.

(4) Each polynomial has finitely many roots, hence the roots of degree n are countable.

(5) $\mathbb{A}$ is countable as the countable union of these roots. $\square$

2.2 Construction of the real numbers

Def 2.2.1 : A subset $L \subseteq \mathbb{Q}$ is a cut if we have $L \neq \emptyset$, $L \neq \mathbb{Q}$, L has no largest element, and $x \in L \Rightarrow y \in L \ \forall y < x$.

The set of real numbers $\mathbb{R}$ is the set of all cuts with order relation given by $\subseteq$, addition given by $L + M = \{l + m \mid l \in L, m \in M\}$ and multiplication given by $L \cdot M = \{l \cdot m \mid l \in L, m \in M ; l, m \geq 0\} \cup \{r \in \mathbb{Q} \mid r < 0\}$ for positive real numbers K, M , which can be extended to negative numbers in the natural way.

TH 2.3 : Every non-empty subset of $\mathbb{R}$, that is bounded from above, has a least upper bound.

(1) Let $A \subseteq \mathbb{R}$ be bounded by $M \in \mathbb{R}$.

(2) For each $L \in A$ we have $L \subseteq M$.

(3) Let $\tilde{L} = \bigcup_{L \in A} L$.

(4) Obviously $\tilde{L}$ is nonempty and a proper subset of $\mathbb{Q}$.

(5) If $\tilde{L}$ would have a largest element l , then $l \in L$ for some $L \in A$ and l would have to be largest element of L , a contradiction.

(6) If $r \in \tilde{L}$ and $s \leq r$ then $r \in L$ for some $L \in A$ and $s \in L$ hence $s \in \tilde{L}$.

(7) This means that $\tilde{L}$ is a cut; $\tilde{L} \in \mathbb{R}$.

(8) Moreover $L \subseteq \tilde{L}$ for all $L \in A$ hence $\tilde{L}$ is an upper bound on A .

(9) Also $\tilde{L} \subseteq M$.

(10) Since M is an arbitrary upper bound, $\tilde{L}$ is the least upper bound. $\square$

TH 2.4 : Every real number can be represented by a unique b -adic expansion for $b \geq 2$.

(1) Given a real number as a cut $F \subseteq \mathbb{Q}$ let $a_0 = \max\{n \in \mathbb{N} \mid n \in \mathbb{Q}\}$ and define recursively

$$a_n = \max\{a \in \{0, \dots, b-1\} \mid a_0 + \sum_{i=1}^{n-1} a_i b^i + ab^n \in F\}.$$

(2) By this we get a representation $a_0.a_1a_2a_3\dots$ of F , which has no tail of zeros.

(3) The other way round, given such a representation, it describes a unique cut by

$$F = \bigcup_{n \in \mathbb{N}} \{q \in \mathbb{Q} \mid q \leq a_0 + \sum_{i=1}^n a_i b^i\}.$$

$\square$

2.3 Uncountable sets

TH 2.5 : The real numbers $\mathbb{R}$ are uncountable.

(1) If $\mathbb{R}$ is countable then $[0, 1]$ is countable as well.

(2) Hence there exists a map C from $\mathbb{N}$ onto $[0, 1]$ with

$$C(n) = \sum_{i=1}^{\infty} c_i(n)10^{-i}$$

where $c_i(n) \in \{0, 1, \dots, 9\}$ are the digits in decimal expansion.

(3) Now consider a real number $x = \sum_{i=1}^{\infty} \bar{c}_i 10^{-i} \in [0, 1]$ with $\bar{c}_i \neq c_i(i)$.

(4) Obviously $C(n) \neq x$ for all $n \in \mathbb{N}$. Hence C is not onto. A contradiction. $\square$

TH 2.6 : There is a bijection between $\mathbb{R}$ and $\mathbb{R}^n$.

(1) For simplicity of notation we proof the result for $n = 2$. The general proof use the same idea.

(2) First note that the map $f(x) = \tan(x)$ is a bijection from $(-\pi/2, \pi/2)$ to $\mathbb{R}$ and a linear map from $(-\pi/2, \pi/2)$ to $(0, 1)$ is a bijection.

(3) Hence it is enough if we find a bijection between $(0, 1)^2$ and $(0, 1)$.

(4) Represent a pair of real numbers $(a, b) \in (0, 1)^2$ in decimal expansion $(a, b) = (0.a_1a_2a_3\dots, 0.b_1b_2b_3\dots)$.

(5) Map such a pair to $0.a_1b_1a_2b_2a_3b_3\dots$.

(6) By uniqueness of decimal expansion, using the convention that the expansion has no tail of zeros, this is a bijection. $\square$

2.4 Cardinalities

TH 2.7 : A and the power set $P(A) = \{B \mid B \subseteq A\}$ do not have the same cardinality.

(1) Let $f : A \mapsto P(A)$ be a function and $T := \{x \in A \mid x \notin f(x)\} \in P(A)$.

(2) If f is surjective, there exists $t \in A$ such that $f(t) = T$.

(3) If $t \in T$ we have by the definition of T , $t \notin f(t) = T$.

(4) On the other hand if $t \in T$, we have $t \in T = f(t)$ by the definition. Contradiction.

□

Def 2.8 : $A \trianglelefteq B$: exists injective map from A to B .

TH 2.9 : The relation $A \trianglelefteq B$ (there is an injective map from A to B) defines an order on cardinalities.

(1) Reflexibility and transitivity of the relation are obvious.

(2) We first show antisymmetry : If there exists injections $f : A \mapsto B$ and $g : B \mapsto A$, then there is a bijection.

(3) For a set $S \subseteq A$ let $F(S) := A \setminus g(B \setminus f(S))$ and define $A_0 := \bigcap_{i=0}^{\infty} F^i(A)$.

(4) By the rule of De Morgan and the definition of F we have

$$F\left(\bigcap A_i\right) = \bigcup F(A_i)$$

and hence $F(A_0) = A_0$, which means $g(B \setminus f(A_0)) = A \setminus A_0$.

(5) The map, given by

$$\phi(x) = \begin{cases} f(x) : & x \in A_0 \\ g^{-1}(x) : & x \notin A_0, \end{cases}$$

is a bijection. It remains to show that the order is total :

(6) There is an injective map from A to B or an injective map from B to A . To this end let

$$F := \{f : \bar{A} \mapsto \bar{B} \mid f \text{ is a bijection with } \bar{A} \subseteq A \text{ and } \bar{B} \subseteq B\}.$$

For each chain $C \subseteq F$ we have $\bigcup C \in F$.

(7) Hence by theorem 2.12 below there is an maximal element $(f : \bar{A} \mapsto \bar{B}) \in F$.

(8) We show that $A = \bar{A}$ or $B = \bar{B}$ for this function f .

(9) Suppose neither. Then there exists $a \in A \setminus \bar{A}$ and $b \in B \setminus \bar{B}$.

(10) But now $f \cup \{(a, b)\}$ would be in F . Contradiction to maximality of F . □

TH 2.10 : $\mathbb{R}$ and $P(\mathbb{N})$ have the same cardinality (consequence to previously 2.9).

(1) First we see that $|P(\mathbb{N})| = |\{0, 1\}^{\mathbb{N}}|$ since the map $f(A) = (a_i)$ with $a_i = 1$ for $i \in A$ and $a_i = 0$ for $i \notin A$ is a bijection.

(2) Furthermore we have $|\mathbb{R}| = |\{0, 1\}^{\mathbb{N}}|$ using binary expansions and anti-symmetry of $\preceq$. □

TH 2.11 : The set of $C(\mathbb{R}, \mathbb{R})$ of continuous functions $f : \mathbb{R} \mapsto \mathbb{R}$ has cardinality $|\mathbb{R}|$.

(1) We have $|\mathbb{R}^{\mathbb{Q}}| = |\mathbb{R}^{\mathbb{N}}| = |\{0, 1\}^{\mathbb{N} \times \mathbb{N}}| = |\{0, 1\}^{\mathbb{N}}| = |\mathbb{R}|$ using $|\mathbb{Q}| = |\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$.

(2) But a continuous function on $\mathbb{R}$ is determined by its values on the rational numbers.

(3) Hence $|C(\mathbb{R}, \mathbb{R})| \leq |\mathbb{R}^{\mathbb{Q}}| = |\mathbb{R}|$.

(4) On the other hand the constant functions are continuous hence $|C(\mathbb{R}, \mathbb{R})| \geq |\mathbb{R}|$. □

2.5 The axiom of choice with some consequences

Axiom of choice : For any family of sets $(A_i)_{i \in I}$ with index set I there is a function

$$f : I \mapsto \bigcup_{i \in I} A_i$$

with $f(i) \in A_i$ for all $i \in I$.

Def 2.5.1 : An order is a well-ordering, if every subset of A has a minimal element.

TH 2.12 : Every non-empty partially ordered set, in which every chain (i.e. ordered subset) has an upper bound, contains at least one maximal element.

- (1) Let A be non-empty and partially ordered.
- (2) Assume A has no maximal element.
- (3) Then especially the upper bound of any chain is not maximal.
- (4) We proof that under this assumption there is an unbounded chain.
- (5) Contradiction.
- (6) Let W be the set of all well-ordered chains in A , then by the AC there is a function $c : W \mapsto A$ with $C(K) \in A \setminus K$ and $c(K) > K$.
- (7) Now we call a chain K in A a special chain if K is well ordered and

$$\forall x \in K : C(K_x) = x \text{ where } K_x = \{y \in A \mid y < x\}.$$

(8) We prove that for two special chain K, L either $K = L$ or $K = L_x$ or $L = K_x$ for some $x \in A$, this implies that the union of special chains is a special chain.

- (9) Assume that $K \neq L$ and $K_x \neq L$ for all $x \in A$.

(10) Since K is well ordered $K \setminus L$ has a minimal element k .

(11) Since L is well ordered $L \setminus K_k$ has a minimal element l .

(12) Now $K_k = L_l$ and hence $k = l$, which prove our claim.

(13) Let $\bar{K}$ be the union of all special chains.

(14) This is a special chain and especially a chain and hence bounded by some $a \in A$.

(15) By assumption a is not in A .

(16) Hence there is another $b \in A$ with $b > a$.

(17) Thus $b \notin \bar{K}$.

(18) On the other hand $b \cup \bar{K}$ is a special chain.

(19) Hence $b \cup \bar{K} \subseteq K$ which implies $b \in \bar{K}$, contradiction.

□

TH 2.13 : Every set has a well ordering (application of Zorn's lemma).

(1) Let C be the set of all well ordered subsets of a set with the partial order

$$(C_1, <_1) < (C_2, <_2) : \Longleftrightarrow C_1 \subseteq C_2 \text{ and } <_1 \subseteq <_2 .$$

(2) Every ordered subset of C has an upper bound by the union of elements of the ordered set.

(3) Hence by Zorn's lemma there is a maximum, say $(M, <)$ of C .

(4) Assume $M \neq A$.

(5) Then there is a $x \in A \setminus M$.

(6) But $M \cup \{x\}$ with the definition $M < x$ is well ordered again.

(7) Contradiction to maximality.

(8) Hence $M = A$ is well ordered.

□

TH 2.14 : Every vector space has a basis.

(1) The set Ξ of all linear independent subsets of a vector space V is partially ordered by inclusion $\subseteq$.

(2) Let $\mathbf{C}$ be a chain in Ξ , then $\mathfrak{C} = \bigcup_{C \in \mathbf{C}} C$ is an upper bound of this chain in Ξ .

(3) By Zorn there is a maximal element B in Ξ .

(4) For every $v \in V$ the set $B \cup \{v\}$ is not linear independent, hence v is linear combination of elements in B , and B is a basis of V .

□

Def 2.5.2 : A filter F on a set X is a subset of $P(X)$ such that

$$\begin{aligned} X &\in F, \emptyset \notin F \\ A, B \in F &\Rightarrow A \cap B \in F \\ A \in F, A &\subseteq B \Rightarrow B \in F. \end{aligned}$$

A filter F is maximal if there is no other filter G on X with $F \subset G$ and a filter is an ultra filter if either $A \in F$ or $X \setminus A \in F$ for all $A \subseteq X$.

(*) The existence of an ultrafilter is a further consequence of the axiom of choice.

TH 2.15 : Every filter can be extended to an ultra filter.

(1) Let F be a filter on X and $\mathfrak{F}$ be the set of all filters on X that contain F .

(2) $\mathfrak{F}$ is partially ordered by inclusion.

- (3) If $\mathcal{C}$ is a chain in $\mathfrak{F}$, the set $\bigcup \mathcal{C}$ is upper bound of $\mathcal{C}$ in $\mathfrak{F}$.
- (4) Hence by Zorn there is a maximal filter that contains F .
- (5) It remains to show that a maximal filter is an ultrafilter.
- (6) Assume that a filter F is not ultra.
- (7) Let $Y \subseteq X$ be such that neither $Y \in F$ nor $X \setminus Y \in F$.
- (8) Let $G = F \cup \{Y\}$.
- (9) This set has the intersection property and may obviously be extended to a filter.
- (10) Hence F is not maximal.

□

2.6 The Cauchy functional equation

Def 2.6.1 : A function $f : \mathbb{R} \mapsto \mathbb{R}$ is additive, if it satisfies the Cauchy equation $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. We assume the continuity.

TH 2.16 : Every continuous additive function is linear.

- (1) By additivity of f we have $qf(p/q) = f(p) = pf(1) \Rightarrow f(p/q) = \frac{p}{q}f(1)$ for all $p, q \in \mathbb{N}$.
- (2) Moreover $f(1) = f(0) + f(1)$ hence $f(0) = 0$ and $f(x) = f(0) - f(-x) = -f(-x)$.
- (3) We thus see that f is linear on $\mathbb{Q}$ and by continuity linear on $\mathbb{R}$.

□

TH 2.17 : There are uncountable many nonlinear additive functions.

- (1) Consider $\mathbb{R}$ as a vector space over the field of $\mathbb{Q}$.

(2) By Th 2.12 there is a basis B of this vector space.

(3) This means that any $x \in \mathbb{R}$ has a unique representation of the form $x = \sum_{i=1}^n r_i b_i$ with $\{b_1, \dots, b_n\} \subseteq B$ and $r_i(x) \in \mathbb{Q}$.

(4) Let $f : B \mapsto \mathbb{R}$ be an arbitrary map and define a extension $f : \mathbb{R} \mapsto \mathbb{R}$ by

$$f(x) := \sum_{i=1}^n r_i f(b_i).$$

(5) On the one hand an obvious calculation shows that f is additive.

(6) On the other hand f is linear if and only if f is linear on B .

(7) But there are uncountable many functions $f : B \mapsto \mathbb{R}$ that are not linear.

□

2.7 Ordinal numbers and Goodstein sequences

Def 2.7.1 : An ordinal number is a well order set α with $x = \{y \in \alpha \mid y < x\}$ for all $x \in \alpha$. We define an order on the ordinal numbers by $\alpha < \beta$ if $\alpha \in \beta$.

TH 2.18 : The ordinals are well ordered by $<$.

(1) Let A be a set of ordinals than $\min(A) = \bigcap A$ is well ordered since the elements of A are well ordered.

(2) Moreover A is minimal since $\min(A) \in \alpha$ for all $\alpha \in A$.

□

Def 2.7.2 : Construct ordinals by $\alpha + 1 = \alpha \cup \{\alpha\}$ for each ordinal α and

$$\beta = \bigcup_{\alpha \in A} \alpha$$

for a countable set of ordinals A . Especially

$$0 = \emptyset, \quad n + 1 = n \cup \{n\}, \quad \omega = \bigcup_{i=0}^{\infty} i, \quad (n + 1)\omega = \bigcup_{i=0}^{\infty} n\omega + i,$$

$$\omega^{n+1} = \bigcup_{i=0}^{\infty} i\omega^n, \quad \omega^\omega = \bigcup_{i=0}^{\infty} \omega^i, \dots, \epsilon, \dots$$

$$\begin{aligned} 0 &< 1 < 2 < \dots < \omega < \omega + 1 < \omega + 2 < \dots < \omega + \omega \\ &= 2\omega < 2\omega + 1 < \dots < 3\omega < \dots < \dots < \omega\omega \\ &= \omega^2 < \dots < \omega^3 < \dots < \omega^\omega < \dots < \omega^{\omega^\omega} < \dots < \omega^{\omega^{\omega^{\dots}}} = \epsilon \dots \end{aligned}$$

Def 2.7.3 : Given a number $N \in \mathbb{N}$ we define the Goodstein sequence recursively. $G_2(N) = N$. If $G_n(N)$ is given, write $G_n(N)$ in hereditary base n notation. Replace n by $n + 1$ and subtract one to obtain $G_{n+1}(N)$. Consider $N = 3$:

$$\begin{aligned} G_2 = 3 &= 2^{2^0} + 2^0 \rightarrow G_3 = 3^{3^0} + 3^0 - 1 = 3^{3^0} \rightarrow G_4 = 4^{4^0} - 1 = 4^0 + 4^0 + 4^0 \\ &\rightarrow G_5 = 5^0 + 5^0 \rightarrow G_6 = 6^0 \rightarrow G_7 = 0. \end{aligned}$$

TH 2.19 All Goodstein sequences eventually terminate with zero,

$$\forall N \in \mathbb{N} \exists n \in \mathbb{N} : G_n(N) = 0.$$

(1) We construct a parallel sequence of ordinal numbers not smaller than a given Goodstein sequence by recursion.

(2) If $G_n(N)$ is given in hereditary base n expansion, replace n by the smallest ordinal number ω .

(3) The "base change" operation in the construction of the Goodstein sequence does not change the ordinal number of the parallel sequences, on the other hand subtracting one is decreasing this sequence.

(4) Since the ordinals are well-ordered the parallel sequences terminates with zero and so does the Goodstein sequence.

□